

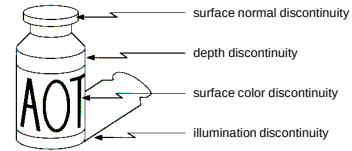
Juergen Gall

Edges and Corners
Computer Vision
MA-INF 2201 / MA-MOROB-M04
WS25/26

Origin of edges

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Edges are caused by a variety of factors:



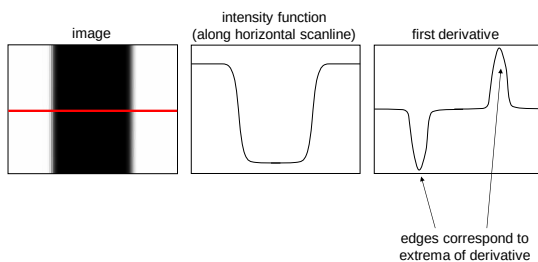
Source: S. Seitz

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Characterizing edges

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- An edge is a place of rapid change in the image intensity function



Source: S. Lantini

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Derivatives with convolution

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For 2D function $f(x,y)$, the partial derivative is:

$$\frac{\partial f(x,y)}{\partial x} = \lim_{\varepsilon \rightarrow 0} \frac{f(x+\varepsilon, y) - f(x, y)}{\varepsilon}$$

For discrete data, we can approximate using finite differences:

$$\frac{\partial f(x,y)}{\partial x} \approx \frac{f(x+1, y) - f(x, y)}{1}$$

To implement above as convolution, what would be the associated filter?

Source: K. Grauman

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Partial derivatives of an image

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Source: S. Lantini

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Finite difference filters

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Other approximations of derivative filters exist:

Prewitt: $M_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$; $M_y = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$

Sobel: $M_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$; $M_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$

Roberts: $M_x = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$; $M_y = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Source: K. Grauman

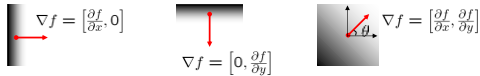
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Image gradient



The gradient of an image:

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$$



- The gradient points in the direction of most rapid increase in intensity
- The gradient direction is given by $\theta = \tan^{-1} \left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x} \right)$
- The edge strength is given by the gradient magnitude

$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

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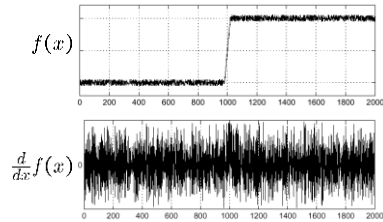
Source: S. Seitz

Effects of noise



Consider a single row or column of the image

- Plotting intensity as a function of position gives a signal



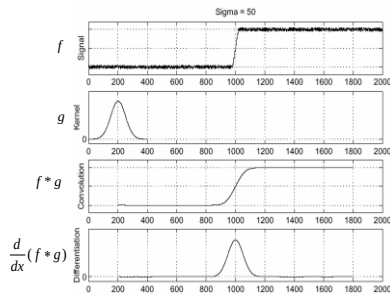
Where is the edge?

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Source: S. Seitz

Solution: smooth first



- To find edges, look for peaks in $\frac{d}{dx}(f * g)$

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Source: S. Seitz

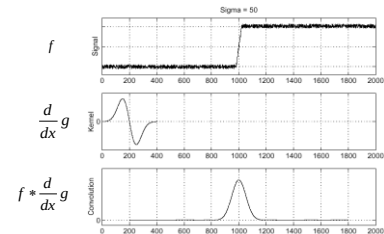
Derivative theorem of convolution



Differentiation is convolution, and convolution is associative:

$$\frac{d}{dx}(f * g) = f * \frac{d}{dx}g$$

This saves us one operation:

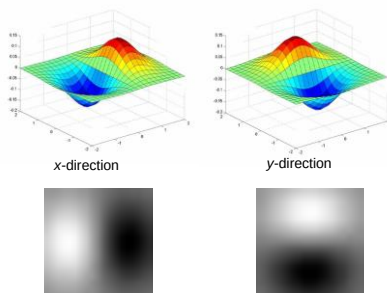


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Source: S. Seitz

Derivative of Gaussian filter



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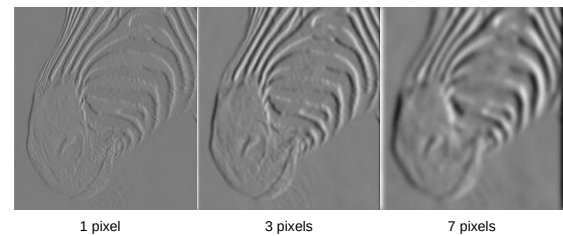
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Source: S. Leibe

Scale of Gaussian derivative filter



Smoothed derivative removes noise, but blurs edge. Also finds edges at different "scales"



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Source: D. Forsyth

Review: Smoothing vs. derivative filters

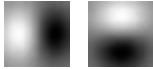
Smoothing filters

- Gaussian: remove "high-frequency" components; "low-pass" filter
- Can the values of a smoothing filter be negative?
- What should the values sum to?
 - **One**: constant regions are not affected by the filter



Derivative filters

- Derivatives of Gaussian
- Can the values of a derivative filter be negative?
- What should the values sum to?
 - **Zero**: no response in constant regions
- High absolute value at points of high contrast



Source: S. Lamberik

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The Canny edge detector



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The Canny edge detector



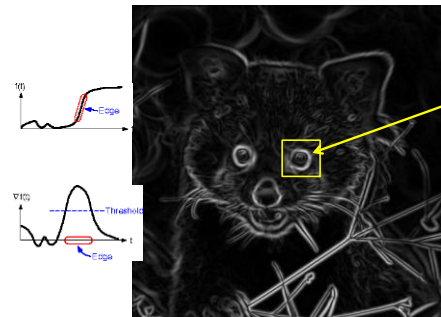
Magnitude of gradient

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The Canny edge detector



How to turn these thick regions of the gradient into curves?

Magnitude of gradient

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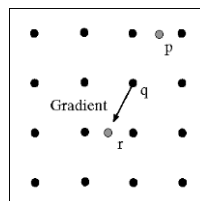
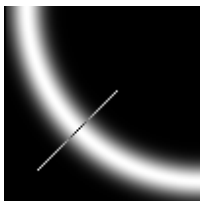
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Non-maximum suppression

Check if pixel is local maximum along gradient direction, select single max across width of the edge

- requires checking interpolated pixels p and r



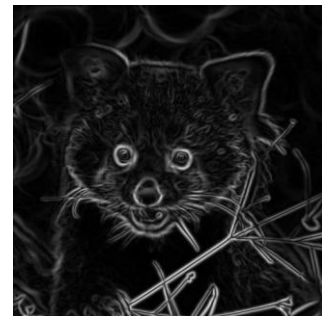
Source: S. Seitz

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The Canny edge detector



Magnitude of gradient

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The Canny edge detector



Non-maximum suppression

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The Canny edge detector



Threshold

Problem:
pixels along
this edge
didn't
survive the
thresholding

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The Canny edge detector



Problem:
Too much
noise

Lower threshold

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Hysteresis thresholding



Use a high threshold to start edge curves, and a low threshold to continue them.



$$t_0 \leq \begin{cases} \|\nabla f(x, y)\| \geq t_1 & \text{definitely an edge} \\ \|\nabla f(x, y)\| < t_1 & \text{maybe an edge, depends on context} \\ \|\nabla f(x, y)\| < t_0 & \text{definitely not an edge} \end{cases}$$

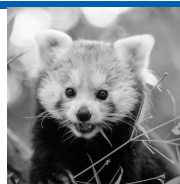
Source: S. Seitz

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Hysteresis thresholding



Low threshold



High threshold



Hysteresis threshold

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Canny edge detector



1. Filter image with derivative of Gaussian
2. Find magnitude and orientation of gradient
3. **Non-maximum suppression:**
 - Thin wide "ridges" down to single pixel width
4. **Linking and thresholding (hysteresis):**
 - Define two thresholds: low and high
 - Use the high threshold to start edge curves and the low threshold to continue them

J. Canny, *A Computational Approach To Edge Detection*, IEEE Trans. Pattern Analysis and Machine Intelligence, 8:679-714, 1986.

Source: S. Seitz

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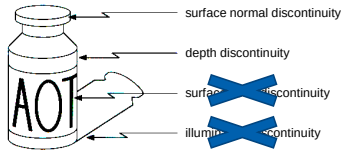
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Relevance of edges

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Application shape matching



Source: S. Seitz

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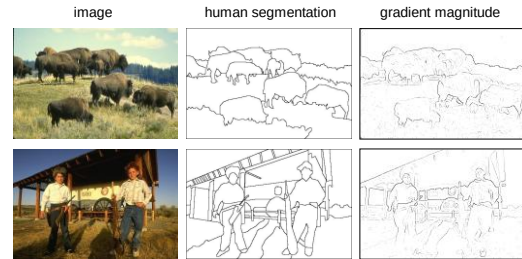
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Not all edges are relevant...

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Berkeley segmentation database:

<http://www.eecs.berkeley.edu/Research/Projects/CS/vision/grouping/resources.html>


Source: S. Lazebnik

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Is boundary detection solved?

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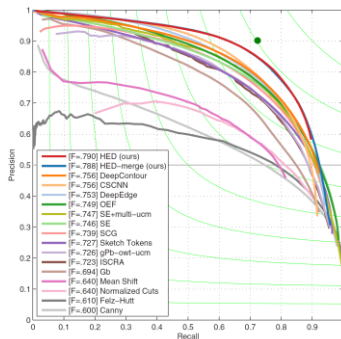
$$\text{Precision} = \frac{tp}{tp + fp}$$

$$\text{Recall} = \frac{tp}{tp + fn}$$

P. Arbelaez et al. **Contour Detection and Hierarchical Image Segmentation**. TPAMI 2011

Saining Xie and Zhuowen Tu **Holistically-Nested Edge Detection**. IJCV 2017

Machine Learning for boundary detection



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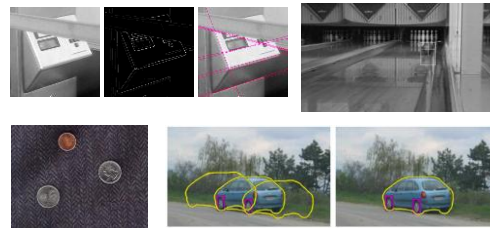
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Fitting

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Want to associate a model with observed features



For example, the model could be a line, a circle, or an arbitrary shape.

Source: K. Grauman

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Chamfer distance

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- Average distance to nearest feature

$$D_{\text{chamfer}}(T, I) \equiv \frac{1}{|T|} \sum_{t \in T} d_I(t)$$



- T: template shape → a set of points
- I: image to search → a set of points
- d(t): min distance for point t to some point in I

Source: K. Grauman

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Chamfer distance

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Average distance to nearest feature

$$D_{\text{chamfer}}(T, I) \equiv \frac{1}{|T|} \sum_{t \in T} d_I(t)$$



Edge image

Can it be implemented as correlation or convolution?

Source: K. Grauman

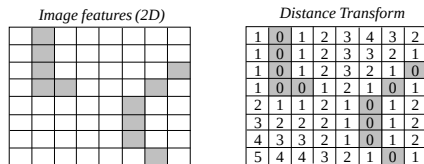
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Distance transform

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Distance Transform is a function $D(\cdot)$ that for each image pixel p assigns a non-negative number $D(p)$ corresponding to distance from p to the nearest feature in the image I

Features could be edge points, foreground points,...

Source: Y. Boykov

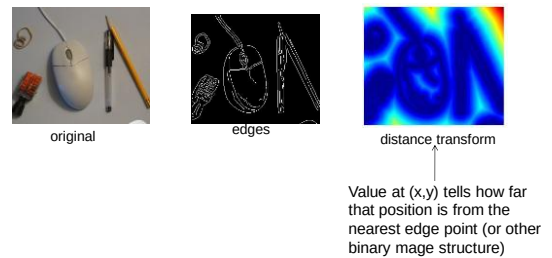
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Distance transform

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Source: K. Grauman

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Distance transform (1D)

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Two pass $O(n)$ algorithm for 1D L_1 norm

1. **Initialize:** For all j
 $D[j] \leftarrow 1_P[j]$ // 0 if j is in P , infinity otherwise

P. Felzenszwalb and D. Huttenlocher. **Distance Transforms of Sampled Functions**. Cornell Computing and Information Science TR2004-1963

Source: D. Huttenlocher

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Distance transform (1D)

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Two pass $O(n)$ algorithm for 1D L_1 norm

1. **Initialize:** For all j
 $D[j] \leftarrow 1_P[j]$ // 0 if j is in P , infinity otherwise
2. **Forward:** For j from 1 up to $n-1$
 $D[j] \leftarrow \min(D[j], D[j-1]+1)$
3. **Backward:** For j from $n-2$ down to 0
 $D[j] \leftarrow \min(D[j], D[j+1]+1)$



P. Felzenszwalb and D. Huttenlocher. **Distance Transforms of Sampled Functions**. Cornell Computing and Information Science TR2004-1963

Source: D. Huttenlocher

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Distance Transform (2D)

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- 2D case analogous to 1D
 - Initialization
 - Forward and backward pass
 - Fwd pass finds closest above and to left
 - Bwd pass finds closest below and to right



Source: D. Huttenlocher

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Distances

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Minkowski distance:

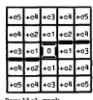
$$d_p(x, y) = \left(\sum_i |x_i - y_i|^p \right)^{1/p}$$

$$d_\infty(x, y) = \lim_{p \rightarrow \infty} \left(\sum_i |x_i - y_i|^p \right)^{1/p} = \max_i (|x_i - y_i|)$$

Special cases: L1-norm, L2-norm (Euclidean)

Approximate Euclidean

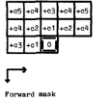
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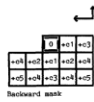
Parallel mask

Forward:
for $i = (\text{size} + 1)/2, \dots, \text{lines do}$
for $j = (\text{size} + 1)/2, \dots, \text{columns do}$

$$v_{i,j} = \min_{(k,l) \in \text{forward mask}} (v_{i+k, j+l} + c(k, l))$$



Forward mask



Backward mask

Backward:
for $i = \text{lines} - (\text{size} - 1)/2, \dots, 1 \text{ do}$
for $j = \text{columns} - (\text{size} - 1)/2, \dots, 1 \text{ do}$

$$v_{i,j} = \min_{(k,l) \in \text{backward mask}} (v_{i+k, j+l} + c(k, l))$$

Chamfer 3-4
 $3 \times \sqrt{2} \approx 4$
 $3 \times 1 = 3$



Chamfer 5-7-11
 $5 \times \sqrt{5} \approx 11$
 $5 \times \sqrt{2} \approx 7$
 $5 \times 1 = 5$

G. Borgefors. **Distance transformations in digital images**. Computer Vision, Graphics, and Image Processing, 34:344–371, 1986.

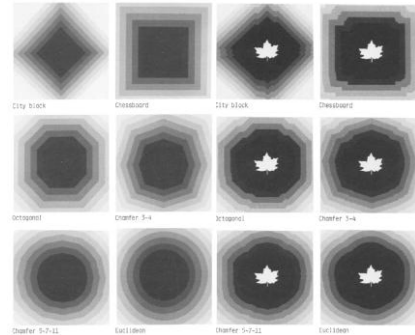
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Approximate Euclidean

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Chamfer distance

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Average distance to nearest feature

$$D_{\text{Chamfer}}(T, I) \equiv \frac{1}{|T|} \sum_{t \in T} d_t(t)$$



Edge image

Distance transform image

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Source: D. Gavrilu

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Chamfer distance

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Edge image

Distance transform image

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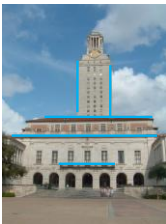
Source: D. Gavrilu

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Line fitting

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- Many objects characterized by presence of straight lines



Lines can be detected by the Hough Transform

Source: R. Gromann

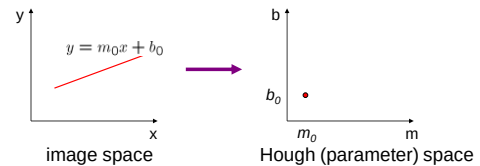
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Hough space

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Connection between image (x, y) and Hough (m, b) spaces:

- A line in the image corresponds to a point in Hough space

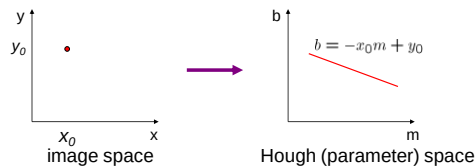
Source: S. Seitz

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Hough space

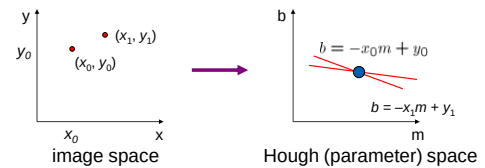


Connection between image (x,y) and Hough (m,b) spaces:

- A line in the image corresponds to a point in Hough space
- What does a point (x_0, y_0) in the image space map to?
- The solutions of $b = -x_0 m + y_0$ (line in Hough space)

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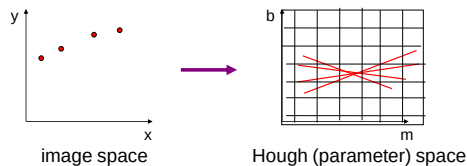
Hough space



- What are the line parameters for the line that contains both (x_0, y_0) and (x_1, y_1) ?
- It is the intersection of the lines $b = -x_0 m + y_0$ and $b = -x_1 m + y_1$

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Hough algorithm



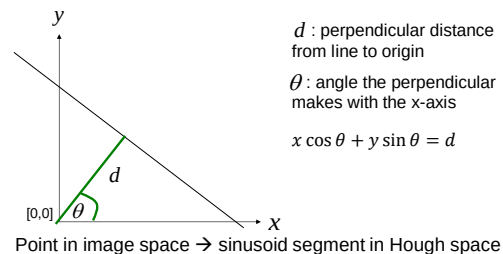
How can we use this to find the most likely parameters (m,b) for the most prominent line in the image space?

- Let each edge point in image space vote for a set of possible parameters in Hough space
- Accumulate votes in discrete set of bins; parameters with the most votes indicate line in image space.

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Polar representation for lines

Issues with usual (m,b) parameter space: can take on infinite values, undefined for vertical lines.



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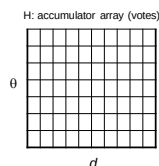
Hough transform algorithm

Using the polar parameterization:

$$x \cos \theta + y \sin \theta = d$$

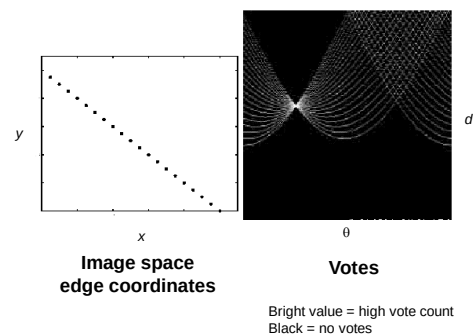
Basic Hough transform algorithm

- Initialize $H[d, \theta] = 0$
- for each edge point $[x,y]$ in the image
for $\theta = 0$ to 180 // some quantization
 $d = x \cos \theta + y \sin \theta$
 $H[d, \theta] += 1$
- Find the value(s) of (d, θ) where $H[d, \theta]$ is maximum
- The detected line in the image is given by
 $d = x \cos \theta + y \sin \theta$



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Lines

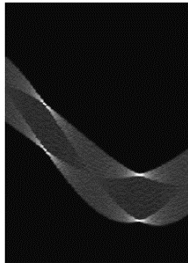


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Lines

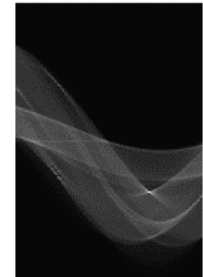
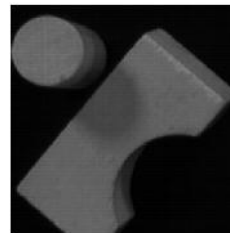


Square :



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Lines

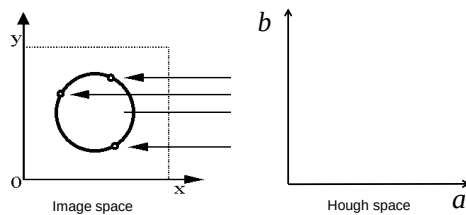


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Hough transform for circles



- Circle: center (a,b) and radius r
 $(x_i - a)^2 + (y_i - b)^2 = r^2$
- For a fixed radius r , unknown gradient direction

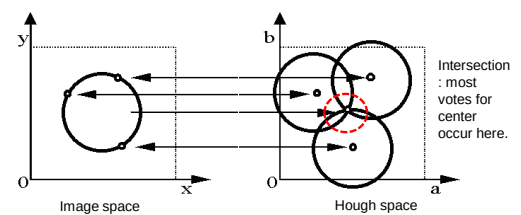


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Hough transform for circles



- Circle: center (a,b) and radius r
 $(x_i - a)^2 + (y_i - b)^2 = r^2$
- For a fixed radius r , unknown gradient direction

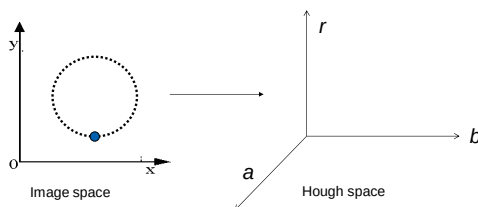


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Hough transform for circles



- Circle: center (a,b) and radius r
 $(x_i - a)^2 + (y_i - b)^2 = r^2$
- For an unknown radius r , unknown gradient direction

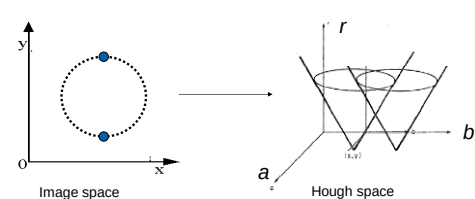


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Hough transform for circles



- Circle: center (a,b) and radius r
 $(x_i - a)^2 + (y_i - b)^2 = r^2$
- For an unknown radius r , unknown gradient direction

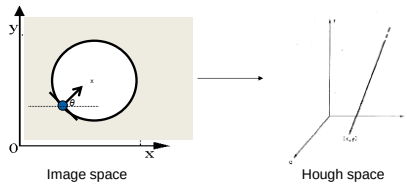


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Hough transform for circles



- Circle: center (a,b) and radius r
 $(x_i - a)^2 + (y_i - b)^2 = r^2$
- For an unknown radius r , **known** gradient direction



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Hough transform for circles



For every edge pixel (x,y) :

For each possible radius value r :

For each possible gradient direction θ :

// or use estimated gradient

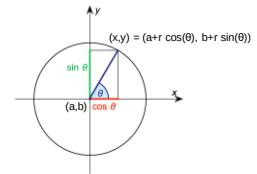
$$a = x - r \cos(\theta)$$

$$b = y - r \sin(\theta)$$

$$H[a,b,r] += 1$$

end

end



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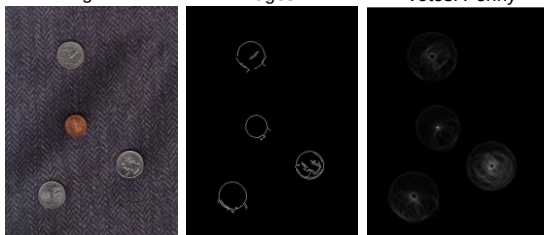
Circles



Original

Edges

Votes: Penny



Note: a different Hough transform (with separate accumulators) was used for each circle radius (quarters vs. penny).

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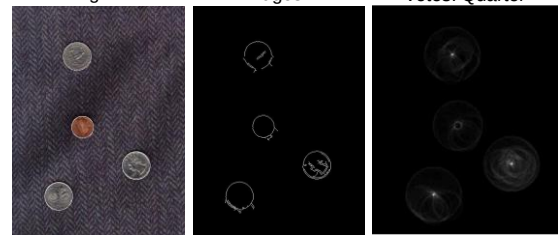
Circles



Original

Edges

Votes: Quarter



Note: a different Hough transform (with separate accumulators) was used for each circle radius (quarters vs. penny).

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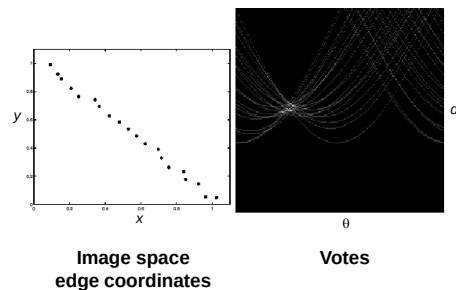
Parameters for analytic curves



Analytic Form	Parameters	Equation
Line	d, θ	$x \cos \theta + y \sin \theta = d$
Circle	x_0, y_0, r	$(x - x_0)^2 + (y - y_0)^2 = r^2$
Parabola	x_0, y_0, p, θ	$(y - y_0)^2 = 4p(x - x_0)$
Ellipse	x_0, y_0, a, b, θ	$(x - x_0)^2/a^2 + (y - y_0)^2/b^2 = 1$

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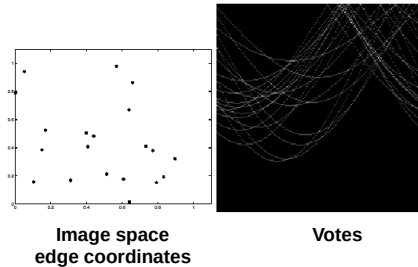
Impact of noise on Hough



What difficulty does this present for an implementation?

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Impact of noise on Hough



Here, everything appears to be "noise", or random edge points, but we still see peaks in the vote space.

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Voting: practical tips



Minimize irrelevant tokens first (take edge points with significant gradient magnitude)

Choose a good grid / discretization

- Too coarse: large votes obtained when too many different lines correspond to a single bucket
- Too fine: miss lines because some points that are not exactly collinear cast votes for different buckets

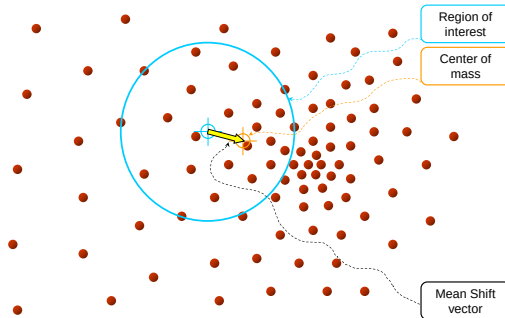
Vote for neighbors also (smoothing in accumulator array) or use mean shift

Utilize direction of edge to reduce free parameters by 1

To read back which points voted for "winning" peaks, keep tags on the votes.

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Mean Shift

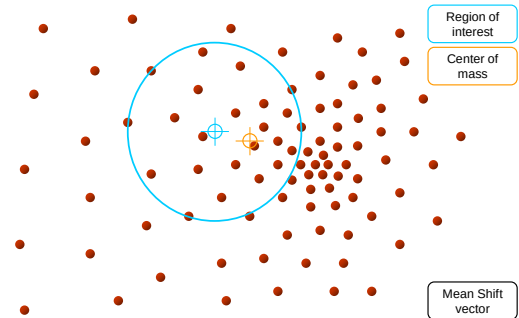


Objective: Find the densest region

Distribution of identical billiard balls

27.10.2025 Jürgen Gall – Institute of Computer Science III – Computer Vision Group Source: Y. Uralovitz and B. Sarel 83

Mean Shift

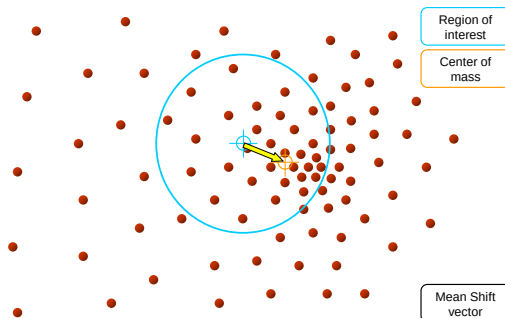


Objective: Find the densest region

Distribution of identical billiard balls

27.10.2025 Jürgen Gall – Institute of Computer Science III – Computer Vision Group Source: Y. Uralovitz and B. Sarel 84

Mean Shift

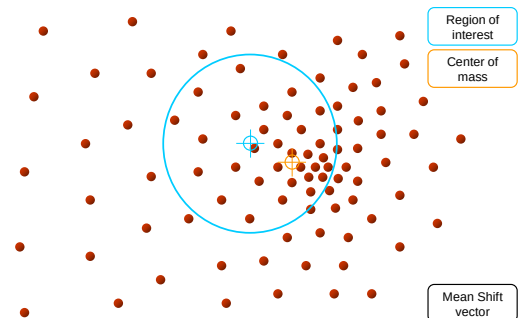


Objective: Find the densest region

Distribution of identical billiard balls

27.10.2025 Jürgen Gall – Institute of Computer Science III – Computer Vision Group Source: Y. Uralovitz and B. Sarel 85

Mean Shift

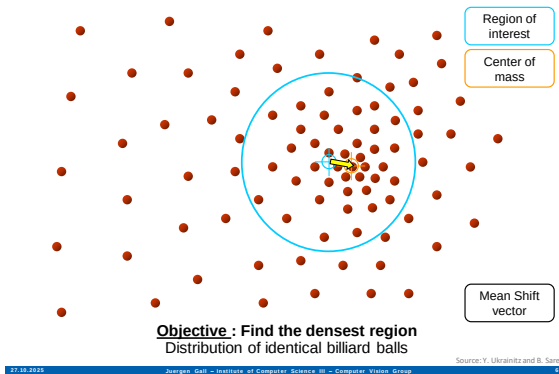


Objective: Find the densest region

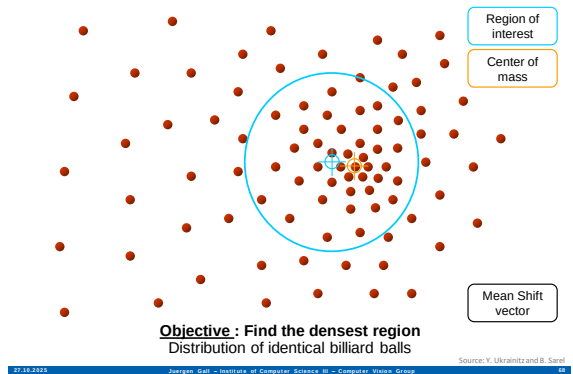
Distribution of identical billiard balls

27.10.2025 Jürgen Gall – Institute of Computer Science III – Computer Vision Group Source: Y. Uralovitz and B. Sarel 86

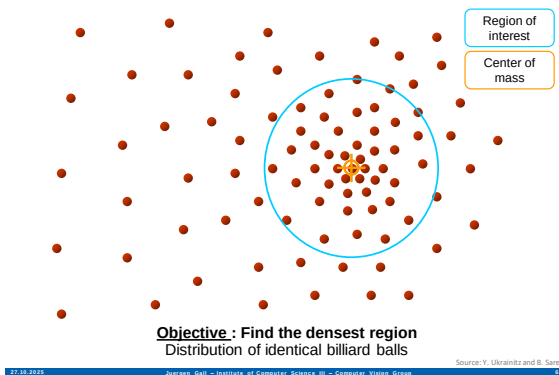
Mean Shift



Mean Shift



Mean Shift



Kernel density estimation

$$P(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n K(\mathbf{x} - \mathbf{x}_i) \quad \text{A function of some finite number of data points } \mathbf{x}_1, \dots, \mathbf{x}_n$$



In practice one uses the forms:

$$K(\mathbf{x}) = c \prod_{i=1}^d k(x_i) \quad \text{or} \quad K(\mathbf{x}) = ck\left(\|\mathbf{x}\|\right)$$

Same function on each dimension

Function of vector length only

Kernels

$$P(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n K(\mathbf{x} - \mathbf{x}_i) \quad \text{A function of some finite number of data points } \mathbf{x}_1, \dots, \mathbf{x}_n$$

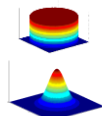


Examples:

• Epanechnikov Kernel $K_E(\mathbf{x}) = \begin{cases} c(1 - \|\mathbf{x}\|^2) & \|\mathbf{x}\| \leq 1 \\ 0 & \text{otherwise} \end{cases}$

• Uniform Kernel $K_U(\mathbf{x}) = \begin{cases} c & \|\mathbf{x}\| \leq 1 \\ 0 & \text{otherwise} \end{cases}$

• Normal Kernel $K_N(\mathbf{x}) = c \cdot \exp\left(-\frac{1}{2} \|\mathbf{x}\|^2\right)$



Gradient ascent

$$\nabla P(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \nabla K(\mathbf{x} - \mathbf{x}_i)$$

Using the Kernel form: $K(\mathbf{x} - \mathbf{x}_i) = ck\left(\left\|\frac{\mathbf{x} - \mathbf{x}_i}{b}\right\|^2\right)$

Size of window

Gradient ascent



$$\begin{aligned}
 \nabla P(x) &= \frac{c}{N} \sum_i \nabla k(\|x - x_i\|^2) \\
 &= \frac{c}{N} \sum_i k'(\|x - x_i\|^2) 2(x - x_i) \\
 &= \frac{2c}{N} \sum_i g(\|x - x_i\|^2) (x_i - x) \quad \text{Use: } g(x) = -k'(x) \\
 &= \frac{2c}{N} \left(\left\{ \sum_i x_i g(\|x - x_i\|^2) \right\} - \left\{ \sum_i g(\|x - x_i\|^2) \right\} x \right) \\
 &= \frac{2c}{N} \sum_i g(\|x - x_i\|^2) \left(\frac{\sum_i x_i g(\|x - x_i\|^2)}{\sum_i g(\|x - x_i\|^2)} - x \right)
 \end{aligned}$$

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Gradient ascent



$$\nabla P(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \nabla K(\mathbf{x} - \mathbf{x}_i)$$

Using the Kernel form: $K(\mathbf{x} - \mathbf{x}_i) = c k\left(\left\|\frac{\mathbf{x} - \mathbf{x}_i}{h}\right\|^2\right)$

We get : Size of window

$$\begin{aligned}
 \nabla P(\mathbf{x}) &= \frac{2c}{N} \sum_i g(\|x - x_i\|^2) \left(\frac{\sum_i x_i g(\|x - x_i\|^2)}{\sum_i g(\|x - x_i\|^2)} - x \right) \\
 g(\mathbf{x}) &= -k'(\mathbf{x})
 \end{aligned}$$

Source: Y. Ukrainitz and B. Sarel

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Mean shift



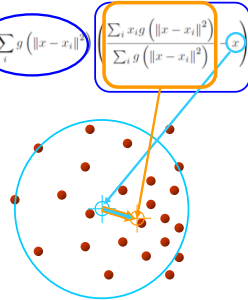
$$\nabla P(\mathbf{x}) = \frac{c}{n} \sum_{i=1}^n \nabla k_i = \frac{2c}{N} \sum_i g(\|x - x_i\|^2) \left(\frac{\sum_i x_i g(\|x - x_i\|^2)}{\sum_i g(\|x - x_i\|^2)} - x \right)$$

Yet another Kernel density estimation !

Simple Mean Shift procedure:
• Compute mean shift vector

$$\mathbf{m}(\mathbf{x}) = \left[\frac{\sum_{i=1}^n \mathbf{x}_i g\left(\frac{\|\mathbf{x} - \mathbf{x}_i\|^2}{h}\right)}{\sum_{i=1}^n g\left(\frac{\|\mathbf{x} - \mathbf{x}_i\|^2}{h}\right)} - \mathbf{x} \right]$$

• Translate the Kernel window by $\mathbf{m}(\mathbf{x})$



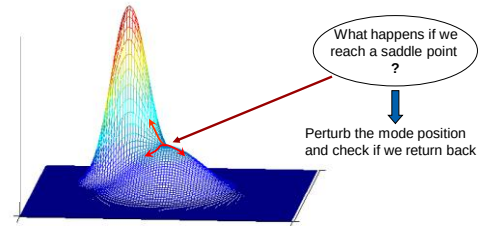
Source: Y. Ukrainitz and B. Sarel

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Mode detection



Updated Mean Shift Procedure:

- Find all modes using the Simple Mean Shift Procedure
- Prune modes by perturbing them (find saddle points and plateaus)
- Prune nearby – take highest mode in the window

Source: Y. Ukrainitz and B. Sarel

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Hough transform: pros and cons



Pros

- All points are processed independently, so can cope with occlusion
- Some robustness to noise: noise points unlikely to contribute consistently to any single bin
- Can detect multiple instances of a model in a single pass

Cons

- Complexity of search time increases exponentially with the number of model parameters
- Non-target shapes can produce spurious peaks in parameter space
- Quantization: hard to pick a good grid size

Source: K. Grauman

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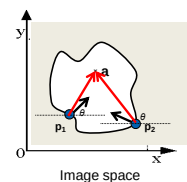
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Generalized Hough transform



What if want to detect arbitrary shapes defined by boundary points and a reference point?



At each boundary point, compute displacement vector: $\mathbf{r} = \mathbf{a} - \mathbf{p}_i$.

For a given model shape: store these vectors in a table indexed by gradient orientation θ .

[D. Ballard, Generalizing the Hough Transform to Detect Arbitrary Shapes, 1980]

Source: K. Grauman

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Generalized Hough transform



To *detect* the model shape in a new image:

For each edge point

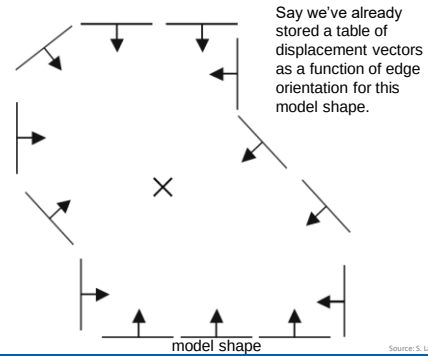
- Index into table with its gradient orientation θ
- Use retrieved r vectors to vote for position of reference point

Peak in this Hough space is reference point with most supporting edges

Assuming translation is the only transformation here, i.e., orientation and scale are fixed.

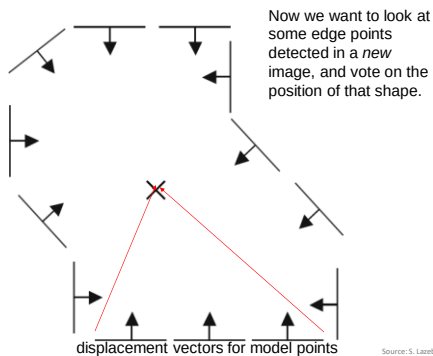
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Example



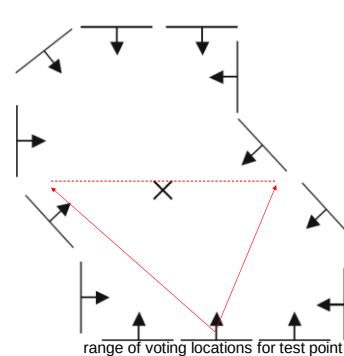
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Example



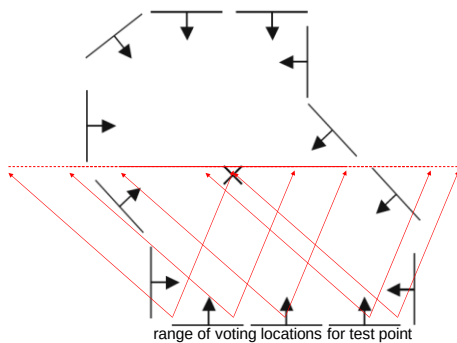
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Example



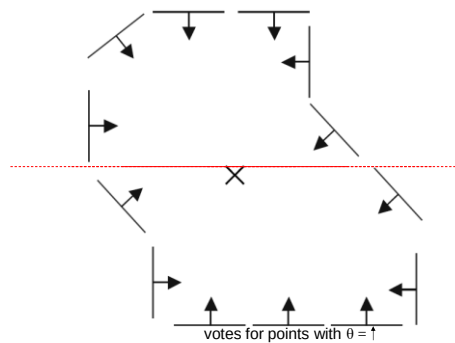
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Example



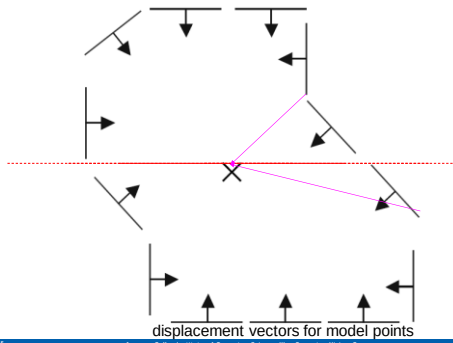
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Example



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Example

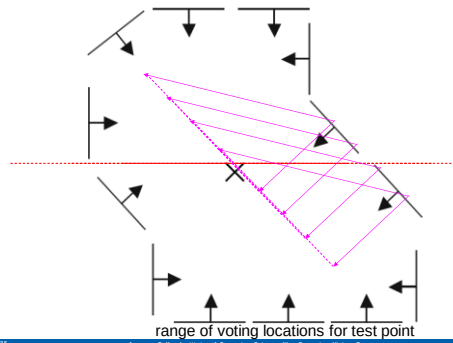


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Source: S. Lazebnik

Example

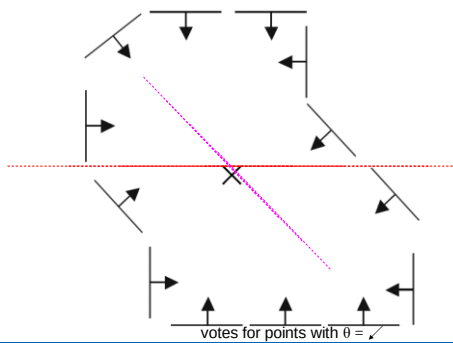


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Source: S. Lazebnik

Example



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Source: S. Lazebnik

Hough voting for object detection



- Parts of an object provide useful spatial information
- Machine Learning: learn how to vote

Gall et al. **Hough Forests for Object Detection, Tracking, and Action Recognition**. IEEE Transactions on Pattern Analysis and Machine Intelligence, Vol. 33, No. 11, 2188-2202, 2011.

27.10.2025

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Hough voting for object detection



Gall et al. **Hough Forests for Object Detection, Tracking, and Action Recognition**. IEEE Transactions on Pattern Analysis and Machine Intelligence, Vol. 33, No. 11, 2188-2202, 2011.

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Source: S. Lazebnik

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